

Big Breeze Coffee Network Proposal

Building Dimensions: 4 Floors — Each 100 ft Wide × 250 ft Deep (25,000 sq ft per floor)

Total Building Area: 100,000 sq ft

1. Network Needs

Big Breeze Coffee has become a popular “social spot” for customers who rely heavily on Wi-Fi. With up to 200 devices connected at once, the existing network cannot support the demand. This proposal outlines a complete network modernization plan to ensure fast and reliable Wi-Fi, secure business operations, better customer satisfaction, protected payment & customer data, reduced downtime, and future growth support.

2. Purpose of the Network Upgrade

Big Breeze Coffee requires a scalable, high-performance network that supports customers, staff systems (POS, payroll), smart coffee equipment, cameras, and security systems.

Benefits include:

- Faster service and happier customers
- Longer customer visits and increased sales
- Reduced downtime
- Strong cybersecurity and data protection
- Support for future expansion

3. Site Survey Summary

A site survey was conducted across all 4 floors (each 100 ft × 250 ft). High-density customer activity was identified on the first two floors, strong concrete/metal framing was noted, and suitable vertical riser and network rack locations were confirmed.

The survey ensures:

- No Wi-Fi dead zones
- Correct number of access points
- Efficient cabling routes
- Cost-effective installation
- Proper backbone placement

4. Internet Bandwidth Requirements

Recommended speeds are:

Internal Backbone (vertical cabling):

- 10 Gbps fiber backbone connecting floors

External Internet Service Provider (Internet

Connection speed):

Standard Business Fiber:

1 Gbps: \$300–\$450/mo

2 Gbps: \$500–\$650/mo

Premium Enterprise Fiber:

1 Gbps: \$600–\$900/mo

2 Gbps: \$900–\$1,300/mo

2 Gbps is recommended to cover peak hour operation

5. Cabling Plan: Backbone & Horizontal Wiring

Vertical Backbone:

- Fiber optic or Cat6e cable for 10 Gbps performance

Horizontal Cabling:

- Cat6/Cat6e wiring to Wireless Access Points, Point of Sale systems, cameras, equipment

* Fiber optics can be seen as better for businesses with a long-term vision for their networks. At the same time, cat6 is identified as a more robust solution for smaller workplaces.

Cabling Cost Estimates:

Cat6: \$0.28–\$0.40/ft + \$1,500–\$2,000 labor

Fiber Optic: \$0.60–\$0.80/ft + \$2,500–\$4,000 labor

6. Hardware Recommendations & Alternatives

Purpose: To lay out recommended devices (routers, switches, Wi-Fi access points) plus lower-cost and higher-end alternatives with prices.

Benefit: Gives the CEO flexibility in budget while still understanding what's "good," "better," and "best," and avoids vendor lock-in by showing multiple options.

A. Router/Firewall:

Ubiquiti UDM-Pro Max: \$599 (recommended)

Netgear PR60X Pro: \$499–\$549 TP-

Link ER707-M2: \$199 (budget)

B. Switches (4 total):

Ubiquiti 2.5G PoE Flex: \$199 each

TP-Link SG2210MP: \$139 each

Netgear GS110EMX: \$179 each

C. Access Points (12 total):

Ubiquiti U7 Pro: \$99 each (recommended)

TP-Link EAP610: \$89 each

Netgear WAX630: \$269 each (premium)

7. Software & Network Management

Purpose: To describe how the network will be monitored, controlled, and secured using management tools and security software.

Benefit: Ensures the network isn't just installed and forgotten, there is a plan to keep it running smoothly, secure, and visible to management.

A. Network Management:

Ubiquiti UniFi Controller: Free

Netgear Insight Pro: \$9.99/device/month

TP-Link Omada SDN: Free–\$8/month

B. Security Software:

Bitdefender GravityZone: \$29–\$35/device/year

Cisco Umbrella: \$3–\$6/user/month

Acronis Backup: \$69–\$89/device/year

8. VLAN Security Architecture

Purpose: To show how the network is logically divided into separate zones (guests, loyalty customers, staff, POS, IoT) for security and performance.

Benefit: Greatly reduces the chances that a hacked guest device or smart gadget can damage critical systems like payment processing, protecting revenue and reputation

Guest Wi-Fi: Isolated, rate-limited

Loyalty Customers VLAN: VPN, priority Wi-Fi access

Staff VLAN: Secure authenticated access, access controls

POS & Finance VLAN: PCI compliant and isolated

IoT & Camera VLAN: Prevents device vulnerabilities

9. Backup & Disaster Recovery

Benefit: To define how data and operations are protected if something goes wrong (hardware failure, outage, cyberattack, accident).

Purpose: Minimizes downtime and data loss so the coffee shop can keep making money even when there's a problem and recover quickly afterward.

Daily cloud backups

Local Network-attached Storage backup

Incremental snapshots every 24 hours

Uninterrupted Power Supply (UPS) battery backup

Secondary Internet Service Provider (ISP) backup

10. Testing & Maintenance Costs

Purpose: To outline the checks done after installation (cable tests, Wi-Fi heatmaps, security tests) and the ongoing maintenance plan.

Benefit: Shows this is not a “set it and forget it” project, there is a plan to verify quality and support performance, so the CEO is not surprised by hidden future costs.

Cable certification: \$300–\$600

Wi-Fi heatmap: \$250–\$450

Penetration testing: \$300–\$500

Annual Maintenance:

Network monitoring: \$400–\$900

Warranty plans: \$200–\$500

Security renewals: \$300–\$900

IT visits: \$600–\$1,500

11. Total Installation Cost

Purpose: To summarize the full cost ranges for low-end, recommended, and premium builds.

Benefit: Gives the CEO a clear financial frame to make a decision that fits the company's budget and growth strategy.

Low-End (Good): \$3,700–\$5,000

High-End (Recommended): \$6,000–\$7,500

Premium: \$10,000–\$13,000

12. Network Deployment Timeline

Phase 1 — Approval & Planning

Length: 1 week

Phase 2 — ISP Installation

Length: 1–2 weeks

Phase 3 — Cabling & Infrastructure

Length: 1 week

Phase 4 — Hardware Installation

Length: 2–3 days

Phase 5 — Configuration & VLAN Setup

Length: 2–3 days

Phase 6 — Testing & Verification

Length: 3–5 days

Phase 7 — Training & Go-Live
Length: 1–2 days

Total Estimated Timeline: 4–6 weeks (up to 8 weeks with ISP delays)

13. Benefits Summary

- Improved customer experience
- Strong security and reliability
- Future-proof design
- 7–10-year hardware lifespan
- Supports 200+ devices with no slowdown
- Easy monitoring and management

14. 5-year Business Growth Projection

- The network upgrade is live
- Big Breeze Coffee runs promotions and incentives (e.g., loyalty rewards, “work-from-café” deals, student discounts, app-based offers)
- Both the better network and promotions together drive more repeat customers, longer stays, and higher average spending

Projections

Year 0 (before upgrade): 100 (baseline revenue / customer index)

Year 1: 112

Year 2: 126

Year 3: 140

Year 4: 155

Year 5: 172

That's roughly 10–12% growth per year, which is realistic for a local business that improves experience + launches smart promos.

13. Final Conclusion

This proposal offers Big Breeze Coffee a modern, secure, scalable, and cost-effective network ready to support high customer traffic and critical business operations across all 4 building floors.